

Overall scientific questions.

Introduction

- Overview of the ISM. How dust forms, why we use CO as a tracer. Why we might use other tracers.
- Jeans collapse a model for star formation.
- IMF / CMF. How the high mass end changes. SF across different environments.
- Quasistatic turbulent core fragmentation as a model for star formation.
- Differences in methods.
- (Nico and Alessio) Literature on decoupling cores from clumps. We would like to know at what stage the gas decouples.
- CHIMPS paper and other extragalactic attempts at this sort of work.

The interstellar medium (ISM) is made up of primarily gas, as well as dust and cosmic rays. The ISM can be divided up based on temperature, ionisation and density, it consists of 5 main phases: the cold molecular medium (CMM), which is defined such that the gas is molecular with a temperature range of $10 - 30\text{ K}$ and a number density between $10^2 - 10^6\text{ cm}^{-3}$; the cold neutral medium (CNM), which is defined such that the gas is mostly atomic with a temperature range of $30 - 100\text{ K}$ and a number density between $1 - 100\text{ cm}^{-3}$; the warm neutral medium (WNM), which is defined such that the gas is mostly atomic with a temperature range of $6000 - 10000\text{ K}$ and number densities below 1 cm^{-3} ; the warm ionised medium (WIM), which is defined such that the gas is mostly ionised with a temperature range of $8000 - 10000\text{ K}$ and number densities around 0.4 cm^{-3} and the hot ionised medium (HIM), which is defined such that the gas is mostly ionised with a temperature range of $10^6 - 10^7\text{ K}$ and number densities below 10^{-2} cm^{-3} (see [Klessen and Glover 2016](#)). The focus of this work will be on the CMM.

Other than primordial elements, every atom formed inside stars. Therefore, initially these elements that make up the gas are hot and have not formed in their molecular state. To get to

the temperature of the CMM, the gas must cool. A star ejects matter throughout its life as stellar winds, as well as at the end of its life in different ways depending on its mass. As the gas moves away from its progenitor the number of ionising photons will decrease which will offset the balance between ionisation and recombination. At these higher temperatures the cooling from Lyman- α becomes important. The general form of the cooling rate of a two level system is shown below.

$$\Lambda_{ul} = \frac{g_u}{g_l} A_{ul} E_{ul} n_l e^{-E_{ul}/k_B T} \quad (1)$$

[Equation 1](#) is the rearranged form of 2.38 in [A. G. Tielens 2005](#). The formula comes about by assuming LTE and that the number density is much less than the critical density. g_u and g_l are statistical weights of the energy states, A_{ul} is the Einstein A coefficient of spontaneous emission, E_{ul} is the energy of the transition and n_l is the number density of the lower state.

$$\Lambda_{HI} = 7.3 \times 10^{-19} \exp\left(\frac{-118400}{T}\right) n_e n_H \quad (2)$$

[Equation 2](#) appears as 2.70 in [A. G. Tielens 2005](#). Above 10^4 K the exponential term stops being negligible which causes the cooling rate to rise rapidly. This alongside the recombination to form atomic hydrogen is what facilitates the transition to the WNM. At higher metallicities the cooling is not dominated by Lyman- α but by other fine structure transitions from heavier elements. Once cooled, [Equation 2](#) shows a ceiling for the temperature in the WNM above which the cooling rate can rapidly exceed the different heating rates [A. G. G. M. Tielens and Hollenbach 1985](#). A different fine structure transition dominates the cooling in the WNM and facilitates the transition to the CNM, the [CII] transition. The energy of this transition is 92 K meaning the exponent in [Equation 1](#) becomes $e^{-92/T}$. This means that below 100 K the cooling from [CII] decreases rapidly. Even though CO is a more efficient coolant at low temperatures with transitional energy of

Project Work

- Motivations (link back to the lit review).
- Methods.
- Conclusions and what that might mean.
- Future work.

Bibliography

- Klessen, Ralf S. and Simon C. O. Glover (Jan. 2016). *Physical Processes in the Interstellar Medium*.
- Tielens, Alexander GGM (2005). *The physics and chemistry of the interstellar medium*. Cambridge University Press.
- Tielens, A. G. G. M. and D. Hollenbach (Apr. 1985). “Photodissociation regions. I. Basic model.” In: ApJ 291, pp. 722–746.